

# NFL Scoring Probabilities – Summary

- **Context:** This project calculates the probability that an offensive drive in the NFL will result in a score using play-by-play data from the 2021–2023 seasons. I will calculate probabilities of scoring from offensive drives using Logistic Regression, Gradient Boosting and Random Forest models. Long-term trends for the team and short-term momentum of the prior drives will also be used to assess the potential success of the drive. Ultimately, I hope to provide a real-time predictive tool that estimates the chances of scoring from a drive, and provides insight for analysts and coaches.
- **Reasoning Behind It:**
  - About 38% of NFL drives from 2021–2023 concluded with a score (TD or FG). Moreover, the likelihood of obtaining a score depends on the nature of the drive (e.g., field position, down, team history). This model can help explain drive situations and trends that affect the probability of a score, creating a more granular way of examining offensive efficiency. In the end, this is a tool to facilitate decision-making during games and in the analytical process.
- **Machine Learning Model:**
  - *Preprocessing:* The missing values in the numerical features were filled using median imputation, and the categorical features were filled with a string 'NA' before one-hot encoding. The features were standardized before any modification to allow for consistent scaling throughout. Any rows with a missing target label ('drive\_ended\_with\_score') were dropped. Plus, more rolling features were created to demonstrate a team's recent momentum.
  - *Feature Importance:* Permutation importance and model-based metrics revealed that the most influential features were: 'posteam\_score\_rate\_rolling', 'drive\_inside20', 'defteam\_score\_rate\_rolling', and 'drive\_first\_downs'. Other important variables included 'play\_number' and 'yardline\_100'.
  - *Different Models Tested:*
    - **Logistic Regression:** Provided a baseline binary classifier for scoring outcomes.
    - **Gradient Boosting:** Built additive models that corrected prior errors and captured non-linear effects.
    - **Random Forest:** Non-linear model that handles complex feature interactions and will provide feature/permutation importance metrics.
- **Model Interpretations**
  - The Random Forest Model predicted scoring probabilities for offensive drives the best in terms of AUC and log loss. It correlated well with the recent offense and defense performance for a team, evidently picking up on trends like persistent success or struggle over the past few drives. While the all the models dealt with short-term momentum and situation context well, they may underemphasize the role of anomalous or highly volatile plays that are not part of a set pattern. It's likely due to the models' preference for steady, replicable indicators—such as drive efficiency and recent scoring streaks—at the cost of singular game-changing plays.
  - **Future Implications:** How would environmental factors – stadium, fans, weather, etc. – affect scoring probabilities? How can integrating player-tracking data and real-time context improve model accuracy?